1B 1 WRITING CHEMICAL EQUATIONS

SPECIFICATION REFERENCE 1.3

LEARNING OBJECTIVES

- Write balanced full and ionic equations, including state symbols, for chemical reactions.

WRITING EQUATIONS: WHAT TO REMEMBER

This topic is a useful summary of what you are expected to do when you write equations.

WRITING FORMULAE FOR NAMES

You may be given the formulae of unfamiliar compounds in a question. You need to work out the formulae of many familiar compounds from their names, and to remember other formulae. You also need to show elements with the correct formulae. It is not possible to provide a complete list, but here are some examples.

You need to remember that:

- oxygen is O_2 and not O
- hydrogen is H_2 and not H
- nitrogen is N_2 and not N
- water is H_2O
- sodium hydroxide is NaOH
- nitric acid is HNO_3.

You should be able to work out that:

- iron(II) sulfate is $FeSO_4$
- iron(III) oxide is Fe_2O_3
- calcium carbonate is $CaCO_3$.

EXAM HINT

Remember to use your Periodic Table in the exam. If you know the formula of *magnesium* sulfate then you can assume *strontium* sulfate has the same formula because Mg and Sr are in the same group.

WRITING AN EQUATION FROM A DESCRIPTION

You will need to convert words into formulae and decide which ones are reactants and which ones are products.

Consider this description: when carbon dioxide reacts with calcium hydroxide, calcium carbonate and water are formed. The wording of the description makes it clear that carbon dioxide and calcium hydroxide are the reactants, and that calcium carbonate and water are the products.

Now you have to write the formulae in the correct places:

$$CO_2 + Ca(OH)_2 \rightarrow CaCO_3 + H_2O$$

The next step is balancing the equation. You need to add up the numbers of all the atoms to make sure that, for each element, the totals are the same on both the left and the right side of the equation. In this example, there is one carbon, one calcium, two hydrogen and four oxygen atoms on each side, so the equation is already balanced.

Here is an example where the first equation you write is not balanced.

The description is:

hydrogen peroxide decomposes to water and oxygen.

The formulae are:

$$H_2O_2 \rightarrow H_2O + O_2$$

This is already balanced for hydrogen, but not for oxygen. With careful practice, which sometimes involves guessing until you get it right, you should be able to write the **coefficients** needed to balance the equation. In this case, the balanced equation is:

$$2H_2O_2 \rightarrow 2H_2O + O_2$$

Most equations are balanced using whole-number coefficients, but using fractions or decimals is usually acceptable. This is especially the case in organic chemistry.

Consider this unbalanced equation for the complete combustion of butane:

$$C_4H_{10} + O_2 \rightarrow CO_2 + H_2O$$

The balanced equation can be either:

$$2C_4H_{10} + 13O_2 \rightarrow 8CO_2 + 10H_2O$$

or

$$C_4H_{10} + 6\tfrac{1}{2}O_2 \rightarrow 4CO_2 + 5H_2O$$

Using 6.5 instead of $6\frac{1}{2}$ is also acceptable.

USING STATE SYMBOLS

Many chemical equations include state symbols. The symbols are:

- (s) = solid
- (l) = liquid
- (g) = gas
- (aq) = aqueous (dissolved in water).

It is important to distinguish between (l) and (aq). A common error is to write $H_2O(aq)$ instead of $H_2O(l)$.

Although it is good practice to include state symbols in all equations, in some cases they are essential, while in other cases they are often omitted. For example, when writing equations to represent ionisation energies in **Topic 2A.4**, it is important to include (g) after each atom and ion. However, in **Topics 4 and 5** you will write equations for organic reactions, where state symbols are often not included.

Here is another description:
when aqueous solutions of silver nitrate and calcium chloride are mixed, a white precipitate of silver chloride forms. As this precipitate settles, a solution of calcium nitrate becomes visible.

After writing the correct formulae, balancing the equation and including state symbols, the equation is:

$$2AgNO_3(aq) + CaCl_2(aq) \rightarrow 2AgCl(s) + Ca(NO_3)_2(aq)$$

ARROWS IN EQUATIONS

Most equations are shown with a conventional (left to right) arrow $\rightarrow$. However, some important reactions are reversible. This means that the reaction can go both in the forward and backward (reverse) directions. The symbol $\rightleftharpoons$ is used in equations for these reactions. You can find guidance later in this book (**Topic 9B**) about when this reversible arrow should be used.

Sometimes a conventional arrow is made longer to allow information about the reaction to be shown above the arrow (and sometimes below it). This information might be about reaction conditions, such as temperature, pressure and the use of a catalyst. In organic chemistry, where reaction schemes are important, a label indicating, for example, Step 1, may be placed on the arrow in a sequence of reactions.

IONIC EQUATIONS

SIMPLIFYING FULL EQUATIONS

Ionic equations show any atoms and molecules involved, but only the ions that react together, and not the **spectator ions**.

This is the easiest method to follow for simplifying equations:

1 Start with the full equation for the reaction.
2 Replace the formulae of ionic compounds by their separate ions.
3 Delete any ions that appear identically on both sides.

WORKED EXAMPLE 1

What is the simplest ionic equation for the neutralisation of sodium hydroxide solution by dilute nitric acid?

The full equation is:

$$NaOH(aq) + HNO_3(aq) \rightarrow NaNO_3(aq) + H_2O(l)$$

You should now consider which of these species are ionic and replace them with ions. In this example, the first three compounds are ionic:

$$Na^+(aq) + OH^-(aq) + H^+(aq) + NO_3^-(aq) \rightarrow Na^+(aq) + NO_3^-(aq) + H_2O(l)$$

After deleting the identical ions, the equation becomes:

$$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$$

WORKED EXAMPLE 2

What is the simplest ionic equation for the reaction that occurs when solutions of lead(II) nitrate and sodium sulfate react together to form a precipitate of lead(II) sulfate and a solution of sodium nitrate?

The full equation is:

$$Pb(NO_3)_2(aq) + Na_2SO_4(aq) \rightarrow PbSO_4(s) + 2NaNO_3(aq)$$

Replacing the appropriate species by ions gives:

$$Pb^{2+}(aq) + 2NO_3^-(aq) + 2Na^+(aq) + SO_4^{2-}(aq) \rightarrow PbSO_4(s) + 2Na^+(aq) + 2NO_3^-(aq)$$

After deleting the identical ions, the equation becomes:

$$Pb^{2+}(aq) + SO_4^{2-}(aq) \rightarrow PbSO_4(s)$$

These remaining ions are not deleted because they are not shown identically. Before the reaction they were free-moving ions in two separate solutions (Pb^{2+} and SO_4^{2-}). After the reaction they are joined together in a solid precipitate ($PbSO_4$).

WORKED EXAMPLE 3

Carbon dioxide reacts with calcium hydroxide solution to form water and a precipitate of calcium carbonate.

The full equation is:

$$CO_2(g) + Ca(OH)_2(aq) \rightarrow CaCO_3(s) + H_2O(l)$$

Replacing the appropriate species by ions gives:

$$CO_2(g) + Ca^{2+}(aq) + 2OH^-(aq) \rightarrow CaCO_3(s) + H_2O(l)$$

Note that carbon dioxide and water are molecules, so their formulae are not changed. In this example, no ions are shown identically on both sides, so this is the simplest ionic equation.

LEARNING TIP

Consider carefully what the correct state symbol for a species should be. It may be different in different reactions. For example, water is never $H_2O(aq)$, but it may be $H_2O(s)$, $H_2O(l)$ or $H_2O(g)$, depending on the temperature.

IONIC HALF-EQUATIONS

We write ionic half-equations for reactions involving oxidation and reduction, and they usually show what happens to only one reactant. A simple example is the reaction that occurs at the negative electrode during the electrolysis of aqueous sulfuric acid:

$$2H^+(aq) + 2e^- \rightarrow H_2(g)$$

You will learn much more about ionic half-equations in **Topic 8**.

CHECKPOINT

1. Sodium thiosulfate ($Na_2S_2O_3$) solution reacts with dilute hydrochloric acid to form a precipitate of sulfur, gaseous sulfur dioxide and a solution of sodium chloride. Write an equation, including state symbols, for this reaction.
2. Solutions of ammonium sulfate and sodium hydroxide are warmed together to form sodium sulfate solution, water and ammonia gas. Write the simplest ionic equation for this reaction.

SUBJECT VOCABULARY

coefficient the technical term for the number written in front of species when balancing an equation

spectator ion an ion that is there both before and after the reaction but is not involved in the reaction

1B 2 TYPICAL REACTIONS OF ACIDS

SPECIFICATION REFERENCE 1.12(ii)

LEARNING OBJECTIVES

- Relate ionic and full equations, with state symbols, to observations from simple test-tube reactions, for reactions of acids.

INTRODUCTION

Acids are common reagents in chemistry. In this topic, we summarise some of their typical reactions, using hydrochloric, nitric, sulfuric and phosphoric acids.

In each of these reactions, a salt is formed. The reactions can be used to prepare samples of salts.

ACIDS WITH METALS

A general equation for these reactions is:

metal + acid → salt + hydrogen

Bubbles of hydrogen gas form, and if the salt formed is soluble, then a solution forms.

The metal must be sufficiently reactive to react in this way. For example, magnesium reacts but copper does not.

Typical equations for magnesium and hydrochloric acid are:

$$Mg + 2HCl \rightarrow MgCl_2 + H_2$$

$$Mg(s) + 2H^+(aq) \rightarrow Mg^{2+}(aq) + H_2(g)$$

These reactions may appear to be examples of neutralisation reactions because the H^+ ions are removed from the solution when they react with the metal. However, as the H^+ ions gain electrons from the metal and are converted to $H_2(g)$, it means that the H^+ ions are reduced, not neutralised.

ACIDS WITH METAL OXIDES AND INSOLUBLE METAL HYDROXIDES

A general equation for these reactions is:

metal oxide + acid → salt + water

metal hydroxide + acid → salt + water

The reactivity of the metal does not matter in these reactions because in the reactant it is present as metal ions, not metal atoms. The only observation is likely to be the formation of a solution.

Typical equations for copper(II) oxide and zinc hydroxide reacting with sulfuric acid are:

$$CuO + H_2SO_4 \rightarrow CuSO_4 + H_2O$$

$$CuO(s) + 2H^+(aq) \rightarrow Cu^{2+}(aq) + H_2O(l)$$

$$Zn(OH)_2 + H_2SO_4 \rightarrow ZnSO_4 + 2H_2O$$

$$Zn(OH)_2(s) + 2H^+(aq) \rightarrow Zn^{2+}(aq) + 2H_2O(l)$$

These reactions can be classified as neutralisation reactions because the H^+ ions react with O^{2-} or OH^- ions. They are not redox reactions because there is no change in the oxidation number of any of the species.

ACIDS WITH ALKALIS

Metal hydroxides that dissolve in water are called alkalis. A general equation for these reactions is:

alkali + acid → salt + water

There are no visible changes during these reactions, although if a thermometer is used, a temperature rise can be noted.

Typical equations for sodium hydroxide reacting with phosphoric acid are:

$$NaOH + H_3PO_4 \rightarrow NaH_2PO_4 + H_2O$$

$$2NaOH + H_3PO_4 \rightarrow Na_2HPO_4 + 2H_2O$$

$$3NaOH + H_3PO_4 \rightarrow Na_3PO_4 + 3H_2O$$

There are three replaceable hydrogens in phosphoric acid. The salt formed depends on the relative amounts of acid and alkali used. The ionic equation for all these reactions is:

$$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$$

These reactions can be classified as neutralisation reactions because the H^+ ions react with OH^- ions. They are not redox reactions because there is no change in the oxidation number of any of the species.

ACIDS WITH CARBONATES

A general equation for these reactions is:

metal carbonate + acid → salt + water + carbon dioxide

Bubbles of carbon dioxide gas form. If the salt formed is soluble, then a solution forms.

Typical equations for lithium carbonate reacting with hydrochloric acid are:

$$Li_2CO_3 + 2HCl \rightarrow 2LiCl + H_2O + CO_2$$

$$CO_3^{2-}(aq) + 2H^+(aq) \rightarrow H_2O(l) + CO_2(g)$$

These reactions can be classified as neutralisation reactions because the H^+ ions react with CO_3^{2-} ions. They are not redox reactions because there is no change in the oxidation number of any of the species.



▲ **fig A** Many old buildings are made from carbonates such as limestone. Centuries of reaction between limestone and acids in the atmosphere have caused damage to the walls of Kolossi Castle in Cyprus.

ACIDS WITH HYDROGENCARBONATES

Hydrogencarbonates are compounds containing the hydrogencarbonate ion (HCO_3^-), and they react with acids in the same way as carbonates. The best-known example is sodium hydrogencarbonate ($NaHCO_3$), commonly known as bicarbonate of soda or baking soda. Baking soda is used in cooking at home and in the food industry. The 'lightness' of baked food such as cakes is due to the formation of bubbles of carbon dioxide in the cake mixture, which cause the cake to rise.

A word equation for the reaction between baking soda and the acid in lemon juice is:

sodium hydrogencarbonate + citric acid → sodium citrate + water + carbon dioxide

A suitable test for the presence of carbonate or hydrogencarbonate ions in a solid or solution is to add an aqueous acid and to test the gas produced with limewater (see **Topic 8B.4**).

LEARNING TIP

Practise writing full and ionic equations for different reactions of acids.

CHECKPOINT

1. Write full equations for the reactions between:

(a) zinc and sulfuric acid

(b) aluminium oxide and hydrochloric acid.

2. Write the simplest ionic equations for the reactions between:

(a) zinc and hydrochloric acid

(b) magnesium carbonate and nitric acid.

1B 3 DISPLACEMENT REACTIONS

SPECIFICATION REFERENCE 1.12(i)

LEARNING OBJECTIVES

- Relate ionic and full equations, with state symbols, to observations from simple test-tube reactions, for displacement reactions.

WHAT IS A DISPLACEMENT REACTION?

As you learn about more chemical reactions, you will know that they are often classified into different types of reaction. You will recognise reaction types such as addition, neutralisation, combustion, oxidation and several others.

In this topic, we look at a reaction type called displacement. In simple terms, it is a reaction in which one element replaces another element in a compound.

DISPLACEMENT REACTIONS INVOLVING METALS

Here are the equations for two **displacement reactions** of metals:

Reaction 1 $Mg(s) + CuSO_4(aq) \rightarrow Cu(s) + MgSO_4(aq)$

Reaction 2 $2Al(s) + Fe_2O_3(s) \rightarrow 2Fe(s) + Al_2O_3(s)$

What do these reactions have in common?

- Both involve one metal reacting with the compound of a different metal.
- Both produce a metal and a different metal compound.
- Both are redox reactions.
- You can see that the metal element on the reactants side has taken the place of the metal in the metal compound on the reactants side.

What are the differences between the reactions?

- Reaction 1 takes place in aqueous solution, but Reaction 2 involves only solids.
- Reaction 1 occurs without the need for energy to be supplied, but Reaction 2 requires a very high temperature to start it.
- Reaction 1 is likely to be done in the laboratory, but Reaction 2 is done for a specific purpose in industry.

METAL DISPLACEMENT REACTIONS IN AQUEOUS SOLUTION

Take a closer look at Reaction 1, shown above. When magnesium metal is added to copper(II) sulfate solution, the blue colour of the solution becomes paler. If an excess of magnesium is added, the solution becomes colourless, as magnesium sulfate forms. The magnesium changes in appearance from silvery to brown as copper forms on it.

The equation can be rewritten as an ionic equation:

$$Mg(s) + Cu^{2+}(aq) + SO_4^{2-}(aq) \rightarrow Cu(s) + Mg^{2+}(aq) + SO_4^{2-}(aq)$$

Cancelling the ions that appear identically on both sides gives:

$$Mg(s) + Cu^{2+}(aq) \rightarrow Cu(s) + Mg^{2+}(aq)$$

Now you can see that this is a redox reaction. Electrons are transferred from magnesium atoms to copper(II) ions, so magnesium atoms are oxidised (loss of electrons) and copper(II) ions are reduced (gain of electrons).

This reaction is just one example of many similar reactions in which a more reactive metal displaces a less reactive metal from one of its salts. You may come across this and other similar reactions as examples used in the measurement of temperature changes.



▲ **fig A** The photo shows what happens when a copper wire is placed in silver nitrate solution for some time. You can see the results of the displacement reaction, the 'growth' on the wire is silver metal and the blue solution contains the copper(II) nitrate that is formed.

METAL DISPLACEMENT REACTIONS IN THE SOLID STATE

Reaction 2 is used in the railway industry to join rails together. You might imagine that a good way to join rails together would be by welding, but the metal rails are good conductors of heat and it is very difficult to get the ends of two rails hot enough for them to melt and join together.

For this reason, the thermite method is used. A mixture of aluminium and iron(III) oxide is positioned just above the place where the two rails are to be joined. A magnesium fuse is lit, and Reaction 2 occurs. It is so exothermic that the iron is formed as a molten metal, which flows into the gap between the two rails. The molten iron cools, joining the rails together.

As in Reaction 1, the equation can be rewritten ionically and simplified:

$$2Al(l) + 2Fe^{3+}(l) + 3O^{2-}(l) \rightarrow 2Fe(l) + 2Al^{3+}(l) + 3O^{2-}(l)$$

then:

$$2Al(l) + 2Fe^{3+}(l) \rightarrow 2Fe(l) + 2Al^{3+}(l)$$

As with Reaction 1, you can see that this is s a redox reaction. Electrons are transferred from aluminium atoms to iron(III) ions, so aluminium atoms are oxidised (loss of electrons) and iron(III) ions are reduced (gain of electrons).



▲ **fig** B The thermite reaction. The flame comes from the highly exothermic reaction forming molten iron. The molten iron will be used to fill the gap between two rails and form a strong join between them.

DISPLACEMENT REACTIONS INVOLVING HALOGENS

In **Topic 8C.2**, we will look at how more reactive halogens can displace less reactive halogens from their compounds. For example, chlorine will displace bromine from a potassium bromide solution. The full, ionic and simplified ionic equations for this reaction are:

$$Cl_2(aq) + 2KBr(aq) \rightarrow Br_2(aq) + 2KCl(aq)$$

$$Cl_2(aq) + 2K^+(aq) + 2Br^-(aq) \rightarrow Br_2(aq) + 2K^+(aq) + 2Cl^-(aq)$$

$$Cl_2(aq) + 2Br^-(aq) \rightarrow Br_2(aq) + 2Cl^-(aq)$$

As with the metal displacement reactions, this is a redox reaction. Electrons are transferred from bromide ions to chlorine, so bromide ions are oxidised (loss of electrons) and chlorine is reduced (gain of electrons).

LEARNING TIP

When describing displacement reactions, be careful to refer to the correct species. For example, in the reaction between magnesium and copper(II) sulfate, magnesium *atoms* and copper(II) *ions* are involved.

CHECKPOINT

1. Iron metal reacts with silver nitrate in a displacement reaction to form silver and iron(II) nitrate. Write a full equation, an ionic equation and a simplified ionic equation for this reaction. Include state symbols in all your equations.
2. A mixture of zinc metal and copper(II) oxide is ignited, causing an exothermic reaction to occur. Write a full equation, an ionic equation and a simplified ionic equation for this reaction. Do not include state symbols in your equations.

SUBJECT VOCABULARY

displacement reaction a reaction in which one element replaces another, less reactive, element in a compound

1B 4 PRECIPITATION REACTIONS

SPECIFICATION REFERENCE
1.12(iii)

LEARNING OBJECTIVES

- Relate ionic and full equations, with state symbols, to observations from simple test-tube reactions, for precipitation reactions.

In this topic, we focus on two aspects of **precipitation reactions**:

- their use in chemical tests
- their use in working out chemical equations.

CHEMICAL TESTS

CARBON DIOXIDE

This may be your earliest memory of a precipitation reaction. When carbon dioxide gas is bubbled through calcium hydroxide solution (often called limewater), a white precipitate of calcium carbonate forms. The relevant equation is:

$$Ca(OH)_2(aq) + CO_2(g) \rightarrow CaCO_3(s) + H_2O(l)$$

The formation of the white precipitate was probably described as the limewater going milky or cloudy.



▲ **fig A** Limewater is a colourless solution. As more carbon dioxide is bubbled through it, the amount of white precipitate increases.

SULFATES

The presence of sulfate ions in solution can be shown by the addition of barium ions (usually from solutions of barium chloride or barium nitrate). The white precipitate that forms is barium sulfate.

For example, when barium chloride solution is added to sodium sulfate solution, the relevant equations are:

$$Na_2SO_4(aq) + BaCl_2(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$$

$$SO_4^{2-}(aq) + Ba^{2+}(aq) \rightarrow BaSO_4(s)$$

This test is covered in more detail in **Topic 8B.4.**

HALIDES

The presence of halide ions in solution can be shown by the addition of silver ions (from silver nitrate solution). The precipitates that form are silver halides.

For example, when silver nitrate solution is added to sodium chloride solution, the relevant equations are:

$$NaCl(aq) + AgNO_3(aq) \rightarrow AgCl(s) + NaNO_3(aq)$$

$$Cl^-(aq) + Ag^+(aq) \rightarrow AgCl(s)$$

This test is covered in more detail in **Topic 8C.4**.

WORKING OUT EQUATIONS

A good example of using a precipitation reaction to work out an equation is the reaction between aqueous solutions of lead nitrate and potassium iodide. Both reactants are colourless solutions. When they are mixed, a yellow precipitate of lead iodide forms.

The word equation for this reaction is:

lead nitrate + potassium iodide → lead iodide + potassium nitrate

Here is an outline of the experiment.

- Place the same volume of a potassium iodide solution in a series of test tubes.
- Add different volumes of a lead nitrate solution to the tubes.
- Place each tube in a centrifuge and spin the tubes for the same length of time.
- Measure the depth of precipitate in each tube.

Table A shows the results of one experiment.

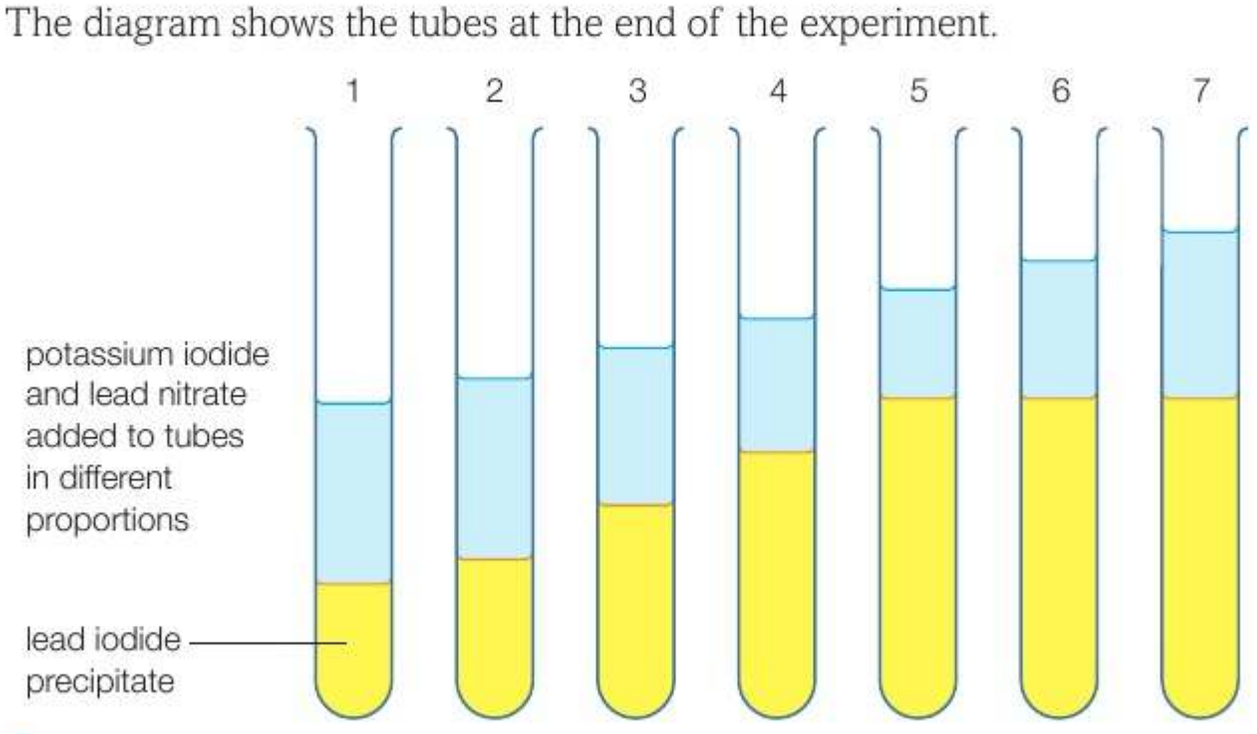
The concentration of both solutions is 1.0 mol dm^{-3}.

The depth of each precipitate indicates the mass of precipitate formed.

TUBE	1	2	3	4	5	6	7
volume of potassium iodide solution / cm^3	5.0	5.0	5.0	5.0	5.0	5.0	5.0
volume of lead nitrate solution / cm^3	0.5	1.0	1.5	2.0	2.5	3.0	3.5
depth of precipitate / cm	2.5	3	4	5	6	6	6

table A Results of the reaction between aqueous solutions of lead nitrate and potassium iodide in one experiment.

The diagram shows the tubes at the end of the experiment.



▲ **fig B** As more lead nitrate solution is added, more lead iodide precipitate forms.

You can see that there is no increase in the amount of precipitate from tube 5 to tube 6. This shows that the reaction is incomplete in tubes 1, 2, 3 and 4, but is complete in tube 5. The amounts, in moles, of reactants used in tube 5 are calculated as follows:

n(potassium iodide) = 0.005 × 1.0 = 0.005 mol

n(lead nitrate) = 0.0025 × 1.0 = 0.0025 mol

This shows that lead nitrate reacts with potassium iodide in the ratio 1 : 2.

The equations for the reaction are:

$Pb(NO_3)_2(aq) + 2KI(aq) \rightarrow PbI_2(s) + 2KNO_3(aq)$

$Pb^{2+}(aq) + 2I^-(aq) \rightarrow PbI_2(s)$

LEARNING TIP

Practise calculating the amounts of reactants and products in tubes 1–4.

CHECKPOINT

1. Write the simplest ionic equation, including state symbols, for:
(a) the test for a sulfate
(b) the test for a chloride.

2. Calculate the amounts, in moles, of each reactant and product in tube 7 in **fig B**.

SKILLS PROBLEM SOLVING

SUBJECT VOCABULARY

precipitation reaction reaction in which an insoluble solid is formed when two solutions are mixed